

ABLEPick

Standard TCP/IP Controller(AT500 Plus)

Data Sheet



V1.0

Updated on 2021/12/01



<http://www.atop.com.tw>

This document is intended to provide customers with brief descriptions of ABLEPick Standard TCP/IP controller(AT500 Plus) and to assist customers to get started quickly.

AT500 Plus is the standard TCP/IP controller of ABLEPick Pick-to-light system. AT500 TCP/IP controller is a data transmission medium between picking control PC/NB and all the pick to lights devices. Which is Ethernet architecture product, following up the standard TCP/IP communication protocol to communicate with the host PC. Each controller has 4 output channels to connect to the pick to light devices, each channel can connect to maximum 30 devices under 30 meters. So one TCP/IP controller can connect to maximum 120 pcs of picking devices.

Packaging

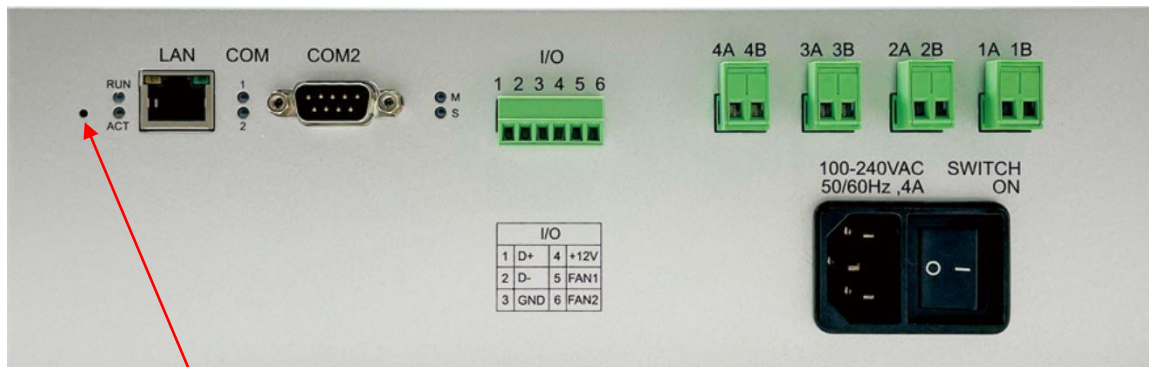
Please check your package contains the following items

- AT500 Plus TCP/IP controller x 1
- Terminal block 2Pin x 4
- Terminal block 6Pin x 1
- Wall mounting brackets and screws x one pair

Hardware specification:

- LAN interface IEEE 802.3 Base band
10/100 Mbps Fast Ethernet,, RJ45 Phone-jack connector
- Output Channel TB2(Lockable), 4 output channels to picking devices.(port1)
- COM Reserved (default)
Option : RS232 or RS485 (ATOP proprietary)
- I/O RS485 (ATOP proprietary) , 12V DC and Fan control.
- Power Input Voltage and current: AC 100V - 240 V, 50/60 Hz, 4A
- Power consumption Maximum 350 Watt
- Power switch AC plug and power switch
- Indicator RUN/ACT/ COM/M/S
- Buzzer Sound warning when rebooting
- Fan Smart fan design
- Operating temperature. 0 °C to 50 °C
- Storage temperature. -20 °C to 70 °C
- Humidity 90% non-condensing
- Dimension 265(L)*136.9(W)*95.2(H) mm
- Weight 2.4kg

Faceplate description - Interface



1. Reset to factory default. Inset pin to push and hold for 3 seconds, then AT500 Plus will reset itself.
2. LAN : RJ45 Ethernet 10/100 Mbps
3. COM : Reserved (default)

Option 1 : RS485 ATOP proprietary protocol to controlling ATOP's old generation pick to light.
RS485 D-type 9 pin male connector

pin	definition.
1	-
2	GND
3	DATA1-
4	RTS-
5	-
6	-
7	-
8	DATA+
9	RTS+

Option 2 : RS232 (D-type 9 pin male connector)

pin	definition.
1	
2	RXD
3	TXD
4	
5	GND
6	
7	RTS
8	CTS
9	VCC(+5V), max 200mA

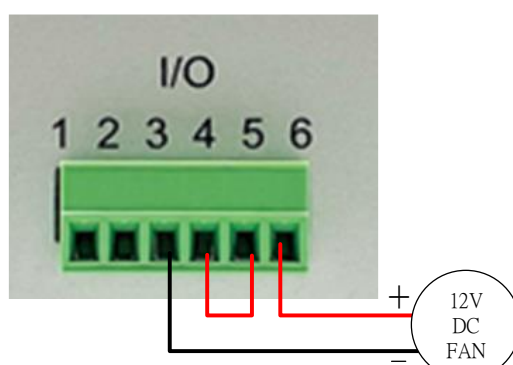
5. I/O : RS485 ATOP proprietary protocol, 12V DC, and Fan Control

pin	definition
1	D+
2	D-
3	GND

4	+12V /5A
5	Dry contact for External Fan control. witch, when environment temperature is igher than 45 °C, it will be trigger
6	

Pin 5 & Pin 6 can be used to control external fan.
Fan control relay : 42°C OFF / 45°C ON ,

DC fan connection method.



The total power consumption of all the connected loading to AT500 plus can not be more than 300 W

6. 4 Output Channels : These 4 output channels 1A1B, 2A2B ~ 4A4B are used to connect to ATOP pick to light devices via twin-wired cable. Each output channel can be connected up to 30 lights under 30 meters in maximum . And please note that the maximum power consumption for each channel is less than 60W.
7. AC plug and Power switch :

Beep sound

The correct power on beep sound :

Beep



LED Indicator

LAN

Message	Description
Amber LED solid on	The connection state is 100M bps
Amber LED is OFF	The connection state is 10M bps
Green blinking	Indicate the connection activity

RUN

Message	Description
LED on/blinking	AT500 plus is running

ACT

Message	Description
LED flashing	data is transmitting on Ethernet

COM / 1

Message	Description
LED Off	AT500 Plus port 1 is not working
LED blinking	AT500 plus port 1 is working

COM / 2

Message	Description
LED Off	AT500 Plus port 2 (COM2) is not working
LED blinking	AT500 plus port 2 (COM2) is working

M

Message	Description
LED on/blinking	AT500 is polling the light modules under its connection

S

Message	Description
LED on/blinking	Data is transmitting back from picking device to AT500

Smart FAN control

AT500 Plus adopt fanless power supply, but it still have an extra internal fan to assist the machine body to dissipate heat to avoid overheating.

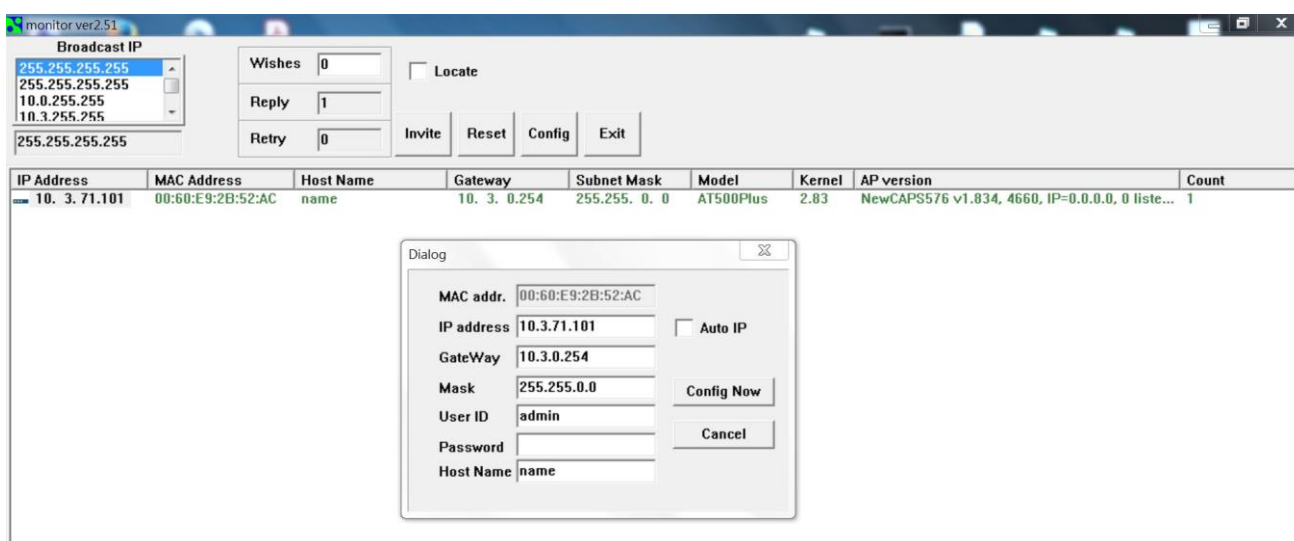
- 1) When AT500 plus reboot, the internal fan will start working by full rotating speed for a while, if the internal temperature of the machine body is less than 40°C, fan will reduce its rotating speed to be the lowest one to last for 5 minutes, then stop.
- 2) When AT500 plus detect the internal temperature of the machine body is over 45 °C, this internal fan will start to work until the internal temperature is low down to 40 °C, then it will reduce its rotating speed, if the internal temperature keeps dropping down, the fan will stop.
- 3) AT500 Plus allow to use pin 5 & Pin 6 of the I/O to execute the external fan control. When the external temperature is over 45 °C, it will trigger the external fan to work. Once the external temperature is low down to 42 °C, it will stop.
- 4) If the external temperature keeps to be higher more than 50 °C, it will trigger beep for sound warning.

Installation Procedures :

Step 1: Connect AT500 Plus via power cord to get AC power input 110 ~ 240VAC.

Step 2: Connect AT500 Plus to one Ethernet network. Using a standard straight-through Ethernet cable when connecting to a Ethernet Switch, or can connect it to PC's Ethernet port via a cross-over Ethernet cable as well. However, It is also necessary to make sure the control PC is within the same network domain as AT500 Plus.

Step 3: AT500 Plus' default IP address is "10.0.50.100" and sub-mask is "255.255.0.0". You can use ATOP's tool "MONITOR.exe" to know and re-configure each its IP address and sub-mask into an appropriate one to let it within the same domain network as control PC.



Step 4: Since AT500 Plus has no DHCP function, so its host control PC/NB needs to assign one IP address which have to be within the same domain as the AT500 Plus. Then you can create one TCP connection to connect it. AT500 Plus's TCP port is 4660

Step 5: For the 4 output channels connection. Please refer the installation guide for the detail installation on the devices side.

Dimension & Drawing

